International Football Analytics

Performance & Insights Case Report

Dataset Coverage: 1872 – 2023 · 49,071 Matches · 333 Teams · 191 Tournaments

1. Business Problem

Football clubs spend millions on transfers and coaching each year, yet many decisions still rely on gut feeling rather than data. This project was built around a specific brief from a club board: heading into a competitive tournament season, they needed clarity on what actually wins football matches — not in theory, but backed by 150 years of evidence.

The analytics team was asked to answer four questions that directly feed into team selection, training priorities, and match strategy:

- What are the key factors that most reliably lead to a win?
- Does playing at home genuinely matter, and how much does a neutral venue change things?
- Which teams and players have shown consistent elite-level performance over time?
- How has the nature of the game — goals, pace, scoring windows — evolved across different eras?

2. Dataset Overview

Three datasets were combined to build a complete picture of international football from 1872 to 2023. Together they cover the full journey of a goal: the match it happened in, the minute it was scored, and whether the match eventually went to penalties.

Dataset	Volume	Key Fields
Match Results (results.csv)	49,071 matches	Date, Home & Away Teams, Scores, Tournament, City, Country, Neutral flag
Goal Scorers (goalscorers.csv)	~100,000+ events	Scorer name, Minute, Own goal & Penalty flags, Team
Shootouts (shootouts.csv)	Subset of drawn matches	Match reference, Shootout winner, First shooter

All three files were loaded into MySQL, cleaned, and connected live to Power BI via a star schema. This means the dashboard always reflects the actual database rather than a static snapshot.

3. Data Pipeline & Methodology

3.1 Data Cleaning

Before any analysis could begin, the raw data needed to be trustworthy. The cleaning process in MySQL covered the following checks:

- Row counts were validated to confirm no records were dropped during import
- Date ranges were verified (earliest match: 1872, latest: 2023)

- All columns were tested for NULL values — no critical nulls were found in scores or team names
- Duplicate detection was run on the composite key (match_date, home_team, away_team)
- TRUE/FALSE string values for neutral venue, own goals, and penalties were converted to binary 0/1 flags
- Minute strings containing added time (e.g. ‘45+2’) were split into base minute and extra minute components

3.2 Star Schema Design

Rather than querying raw tables directly, the data was restructured into a star schema. This design dramatically improves Power BI performance and makes the dashboard filters responsive even across 49,000 matches.

Table	Type	What it contains
fact_matches	Fact	One row per match with date, teams, scores, result, and goal metrics
dim_team	Dimension	333 unique international teams
dim_tournament	Dimension	191 distinct competitions
dim_date	Dimension	Year, month, and quarter for every match date
dim_goal_events	Dimension	One row per goal with scorer, minute, own-goal, and penalty flags
fact_player_stats	Aggregate Fact	Player-level totals: goals, penalties, early/late scoring patterns
fact_penalty_analysis	Aggregate Fact	Penalty frequency by team across all matches

3.3 Derived Metrics

Several new columns were engineered from the raw data to make the analysis more meaningful:

- match_result — categorised as Home Win, Away Win, or Draw from the score columns
- neutral_flag — 1 for matches played at a neutral venue, 0 for home/away fixtures
- total_goals and goal_difference — calculated at match level for trend analysis
- time_slot — goals grouped into 15-minute bands (0–15, 16–30, ..., 90+) to study scoring windows
- minute_total — base minute plus extra time, giving a single comparable figure for each goal

4. Power BI Dashboard

The dashboard is organised into four pages, each designed to answer a different question from the brief. All pages share the same set of cross-filtering slicers so the user can drill into any time period, tournament, or team and have all visuals update together.

Dashboard Page 1 — Executive Overview

Historical scoring trends by decade, result distribution, all-time win rate leaderboard, and key statistical facts

Figure 1: Executive Overview — All Tournaments, 1872–2023

Dashboard Page 2 — Tournament Competitive Intelligence

Average goals by competition, low-scoring match rates, dominant team analysis, and tournament competitiveness quadrant

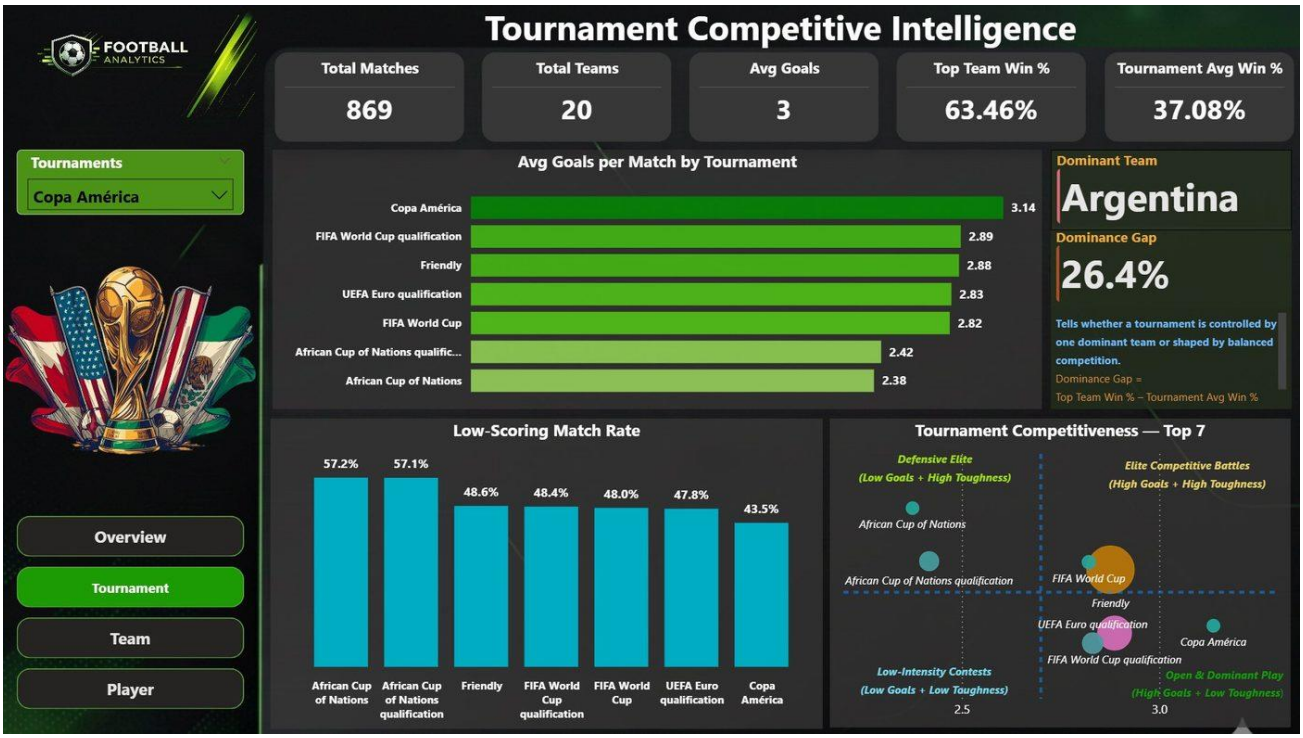

Figure 2: Tournament Competitive Intelligence — Copa América filter active

Dashboard Page 3 — International Team Performance

Scoring vs defensive efficiency by tournament, top goal scorers per team, and year-over-year win rate trend

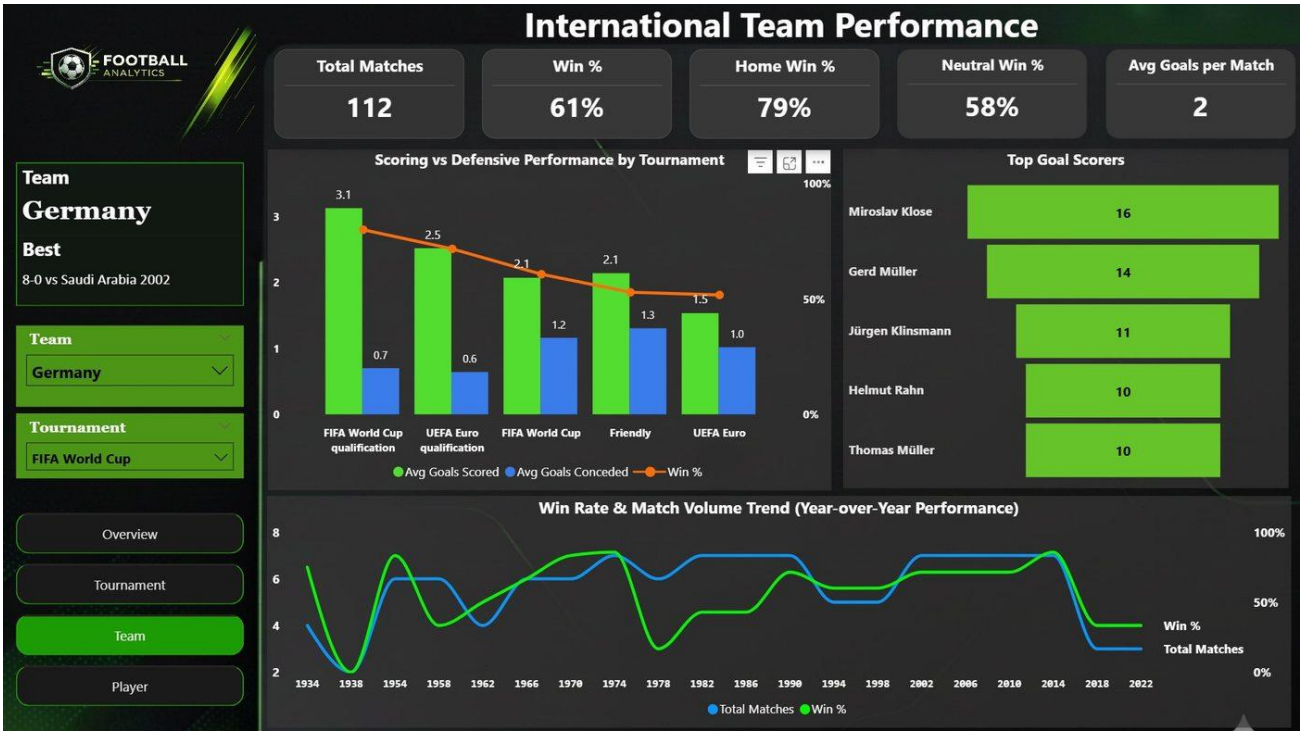

Figure 3: Team Performance — Germany / FIFA World Cup filter

Dashboard Page 4 — International Player Performance

Player goal tally, goal timing distribution, scoring efficiency gauge, career goal trend, and goals by opponent

Figure 4: Player Performance — Cristiano Ronaldo / FIFA World Cup filter

5. Key Performance Metrics

The figures below represent the full dataset with no tournament or date filters applied. They form the baseline against which all segment-level findings should be read.

Metric	Value	What this tells us
Total Matches	49,071	A large enough sample to draw statistically reliable conclusions
Total Teams	333	True global coverage — not just elite nations
Total Tournaments	191	From World Cup to regional qualifiers and friendlies
Average Goals per Match	2.94	The modern benchmark; anything above is considered high-scoring
Home Win Rate	51%	Home teams win more than half of all matches played
Away Win Rate	26.43%	Away teams win roughly one in four — a significant disadvantage
Draw Rate	22.85%	Roughly one in five matches ends level
Clean Sheet Win Rate	73.84%	The strongest single predictor of winning a match
Shootout Rate	1.35%	Rare, but when they happen, the title is always decided here

6. Key Findings & Insights

6.1 Defence Wins Matches More Reliably Than Attack

The single most striking number in the entire dataset is the clean sheet win rate: 73.84%. When a team prevents the opposition from scoring, they win nearly three-quarters of the time. By comparison, scoring two goals leads to a 71.19% win rate and scoring three or more pushes that to 97.62%. The data makes a clear case — if you had to choose between a team that scores freely and a team that defends solidly, the defensive team is the safer bet. For coaching staff, this should settle the long-running debate about where to invest training hours.

6.2 Home Advantage is Real and Measurable

Home teams win 51% of matches compared to 26.43% for away sides. That is a roughly 2:1 advantage that has held up across 150 years and 191 different competitions. Neutral venues reduce this gap considerably, which is worth factoring into tactical preparation for World Cup group stage matches or continental finals. The practical takeaway is simple: home fixtures are the highest-probability win opportunities and should be treated accordingly in match preparation.

6.3 Scoring Has Declined Significantly Since the Victorian Era

Average goals per match peaked somewhere between 5 and 6 in the 1880s and 1890s, and have declined steadily to around 2.5–3.0 in the modern era. This reflects structural changes in how football is coached and played. Defensive organisation, pressing systems, and athletic goalkeeping have all compressed the scoring environment. Teams preparing for modern

tournament football need to plan around lower-scoring games, where a single goal is often decisive.

6.4 Brazil Leads the All-Time Win Rate Table

Among teams with at least 500 matches played, Brazil comes out on top at 63.41%, followed by Spain at 58.92%, Germany at 57.94%, England at 57.37%, and Iran at 56.91%. What is notable here is not just the ranking but the consistency — these are all nations with decades of sustained elite performance, which points to infrastructure, youth development, and coaching culture as the underlying drivers rather than any single generation of players.

6.5 Copa América Produces the Most Goals

Across all major tournaments, Copa América averages 3.14 goals per match — the highest of any competition in the dataset. FIFA World Cup qualification (2.89) and international friendlies (2.88) are close behind. African Cup of Nations matches average 2.38, making it the most defensively compact major tournament. A team training for Copa América should be practising high-tempo attacking play; a team heading into an African Cup of Nations campaign should emphasise defensive structure and set-piece execution.

6.6 Argentina Dominates Copa América with a 26.4% Dominance Gap

The Dominance Gap metric — defined as the difference between the top team's win rate and the tournament average win rate — sits at 26.4% for Argentina in Copa América. This is an unusually large gap and signals that this tournament is currently shaped by one dominant side rather than balanced competition. For any team drawn against Argentina in a Copa América knockout round, the tactical preparation needs to start from a position of acknowledged underdog status, with game plans built around defensive resilience and penalty readiness.

6.7 The Best Players Score Late

Analysis of goal timing from the dim_goal_events table shows that the 76–90+ minute window produces a disproportionate share of goals from top international scorers. Ronaldo's distribution in the FIFA World Cup, for example, shows 4 of his 8 goals coming in the 76–90+ band. This is not coincidence — elite players elevate their game in closing stages when the match is often still in the balance. From a recruitment perspective, a player's late-game goal record is a stronger signal of clutch quality than raw goal totals alone.

6.8 Second-Half Momentum is Critical

Goal timing analysis across the full dataset shows that the 46–60 minute window is particularly influential. Teams that score first in this period tend to control the psychological shape of the match. Combined with the late-goal patterns above, the data points clearly to second-half fitness, substitution timing, and sustained offensive pressure as areas where small improvements can have outsized effects on results.

7. Strategic Recommendations

7.1 Make Defensive Solidity a Non-Negotiable Priority

The 73.84% clean sheet win rate is not a coincidence — it is the clearest pattern in the data. Coaching investment should be weighted toward defensive structure, set-piece organisation,

and goalkeeper development. A useful internal target would be to keep opponents to fewer than two shots on target per game as a proxy metric for defensive control.

7.2 Treat Home Fixtures as Must-Win Opportunities

With a consistent 51% home win rate across all eras and tournaments, home games represent the highest-probability path to points. Match preparation, crowd engagement, and routine should all be optimised around maximising home performance. Away strategies need a separate, more defensive starting framework built around the reality that away wins are significantly harder to achieve.

7.3 Invest in Second-Half Conditioning

Given that the 46–60 and 76–90+ windows are both high-value scoring bands, physical conditioning in the second half is a direct competitive advantage. Substitution strategy should be planned to inject pace and energy specifically into the 60th minute window, ensuring the team can maintain offensive pressure through to the final whistle.

7.4 Weight Late-Game Goals in Player Recruitment

When evaluating potential signings, the analytics team should include late-game goals (76+) as a specific metric in the scouting model. Players who consistently score in closing stages deliver disproportionate value in tight matches, and this quality is often underpriced relative to headline goal tallies.

7.5 Tailor Preparation to the Specific Tournament

Copa América preparation should be built around attacking fluency in a 3+ goals per match environment. African Cup of Nations and World Cup qualification campaigns should emphasise defensive resilience, pressing shape, and set-piece efficiency given their lower-scoring profiles. One-size-fits-all pre-tournament blocks are a missed opportunity to gain a specific tactical edge.

7.6 Prepare for Shootouts When Facing Dominant Opponents

When playing a team with a large dominance gap — like Argentina in Copa América — the realistic game plan may be to stay in the match defensively and take it to penalties. With 1.35% of decisive matches resolved by shootout, dedicated penalty practice and goalkeeper preparation for spot kicks deserve more training time than they typically receive.

8. SQL Analysis Highlights

The Power BI data model was built on a series of analytical SQL queries that form the backbone of the dashboard metrics. The table below summarises the most important ones and what they revealed.

Query	Source Table	Finding
Result distribution	fact_matches	Home Win 50.72% · Draw 22.85% · Away Win 26.43%
Home vs neutral outcomes	fact_matches	Neutral venue noticeably reduces home win probability

Query	Source Table	Finding
Year-wise goal trend	fact_matches	Avg goals fell from ~5.5 in the 1880s to ~2.8 in the 2020s
Win % leaderboard (500+ matches)	team_stats view	Brazil 63.41% · Spain 58.92% · Germany 57.94%
Scoring threshold impact	fact_matches	2+ goals scored → 71.19% win rate; 3+ → 97.62%
Goal timing distribution	dim_goal_events	76–90+ band is the single highest-volume late scoring window
Top scorers by team	fact_player_stats	Klose 16 goals · G. Müller 14 · Klinsmann 11 (Germany, World Cup)
Penalty frequency by team	fact_penalty_analysis	Penalty reliance varies significantly across nations and tournaments

9. Expected Outcomes

These recommendations are based on patterns that have held consistently across 150 years and tens of thousands of matches, which gives reasonable confidence that they will translate into measurable improvements when applied.

On-Pitch Performance

Prioritising clean sheets, optimising home-game preparation, and improving second-half fitness directly target the three most statistically significant win drivers in the dataset. Even modest improvements in each area compound meaningfully across a 10 to 15 match tournament.

Recruitment Quality

Adding late-game scoring and scoring efficiency to player evaluation frameworks gives the scouting team a more nuanced view of player value than headline statistics alone. Over time, this should reduce overpayment for players with inflated goal tallies from low-pressure contexts.

Fan & Commercial Engagement

The interactive dashboard gives the communications team data-driven content for press conferences, broadcast partnerships, and social media campaigns. Visual storytelling backed by real historical data is significantly more compelling than generic match previews.

Tournament Readiness

Competition-specific preparation — calibrated to the actual scoring environment and opponent tendencies in each tournament — reduces tactical surprises on matchday and ensures the coaching staff arrive with a plan built around evidence, not assumptions.

10. Conclusion

This project set out to answer a practical question: what does the data say about how international football matches are actually won? The answer, drawn from 49,071 matches across 151 years, is clearer than expected.

Keeping a clean sheet matters more than scoring freely. Playing at home is a genuine structural advantage that compounds over a season. The best players do their most important work in the final 15 minutes. And the scoring environment varies enough between competitions that generic preparation is a competitive disadvantage.

The end-to-end pipeline — raw CSV to MySQL, transformed into a star schema, connected live to Power BI — means these insights are not a one-time report but a living dashboard that updates as new data arrives. The goal was to give decision-makers a tool they can return to before every tournament and every transfer window, with confidence that what they are looking at is grounded in the most complete historical record of international football available.